

LATENCY-AWARE SELECTIVE MULTI-VIEW RETRIEVAL FOR UNSUPERVISED SOCIAL EVENT CLUSTERING

Kunqing Li, Mengge Ai, Ruiteng Yan, Ke Wen, Enying Bao, Lei Wang, Yuejin Guo, and Xi Chen*

Southwest Minzu University, Chengdu, China; *Corresponding author

{1053961104, 1521076843, 1427645033, 840380055}@qq.com

benying2001@163.com, wanglei@swun.edu.cn, 1771062166@qq.com, cx@swun.edu.cn

ABSTRACT

Streaming social event monitoring requires timely event assignments. Existing retrieval-augmented detectors rely on text queries or invoke multiple evidence views unconditionally, causing query drift and long remote-LLM critical paths. We propose MvSED, which treats multi-view evidence as a bounded residual correction. Text retrieval fixes candidate membership, relevance–redundancy selection chooses rerankers, and a frozen label-free gate accepts only consistent repairs of uncertain assignments. On Event2012, MvSED yields mean NMI, AMI, and ARI changes of +0.00023, +0.00005, and +0.00069, while reducing runtime and LLM calls by 11.3% and 1.3%. On Event2018, the corresponding mean changes are +0.0098, +0.0127, and +0.0567, with runtime and calls reduced by 22.3% and 5.9%. Although Event2018 tokens increase by 11.9%, failed attempts fall by 53.7%, exposing a practical latency–token trade-off. Project page and code: ghostlise.github.io/MvSED.

Index Terms— social event monitoring, short-text clustering, retrieval-augmented generation, multi-view retrieval, end-to-end latency

1. INTRODUCTION

Social event detection groups a stream of short social-media messages into evolving real-world events. It supports crisis monitoring and public-opinion analysis, where an accurate cluster delivered late may already describe a stale situation [1, 2]. Moreover, event memory is updated sequentially, so delayed assignments postpone the context available to later blocks. Timeliness is therefore part of the monitoring objective, not merely an implementation statistic. The task remains difficult because messages are sparse, informal, and lexically diverse; local graph structure alone may not associate different reports of the same event [3, 4].

Retrieval-augmented generation (RAG) supplies complementary context from an external memory [5]. RagSEDE applies this principle to unsupervised social event detection and event-memory evolution [6]. A single text query can miss an informative entity, relation, or temporal pattern, whereas

applying every view to every anchor causes query drift and extra inference. For remotely served LLMs, token volume is not a sufficient proxy for elapsed time: serial calls, failures and retries, retrieval, and service delay all lie on the critical path. Spending more tokens in a better-grounded interaction can therefore be reasonable if it avoids slower repeated interactions.

We address this trade-off with MvSED, which targets lower wall-clock latency subject to preserving clustering quality, rather than minimum token count. Multi-view retrieval is a *selective residual correction*: the text view retrieves the only admissible candidate set; only uncertain assignments activate entity, relation, and temporal reranking; and a deterministic gate accepts a repair only under semantic-support and ranking-margin constraints. Confident decisions bypass auxiliary reasoning, while ambiguous anchors spend additional evidence tokens on a bounded repair path.

Our contribution is a latency-aware selective residual design. Text fixes membership; relevance–redundancy selection admits only useful rerankers; and a frozen label-free gate changes only uncertain assignments supported by consistent evidence. With identical anchor units, this design reduces paired runtime by 11.3–22.3%; Event2012 mean metric changes remain below 0.001, while Event2018 shows favorable observed mean changes.

2. RELATED WORK

Social event detection. Early methods model lexical, user, or topic signals [7, 8]; later systems use graphs, incremental learning, and structural entropy [3, 9, 4]. PromptSED uses label-assisted prompt learning and LSED a label-supervised classifier [10, 11], so their published scores are not mixed with our frozen label-free-inference comparison. We include HISEvent [4] as an external unsupervised quality reference. RagSEDE is the controlled baseline because MvSED retains its anchors, detector, LLM, and memory protocol, changing only residual retrieval and repair.

Retrieval and multi-view reasoning. Dense retrieval and RAG provide external evidence for downstream reason-

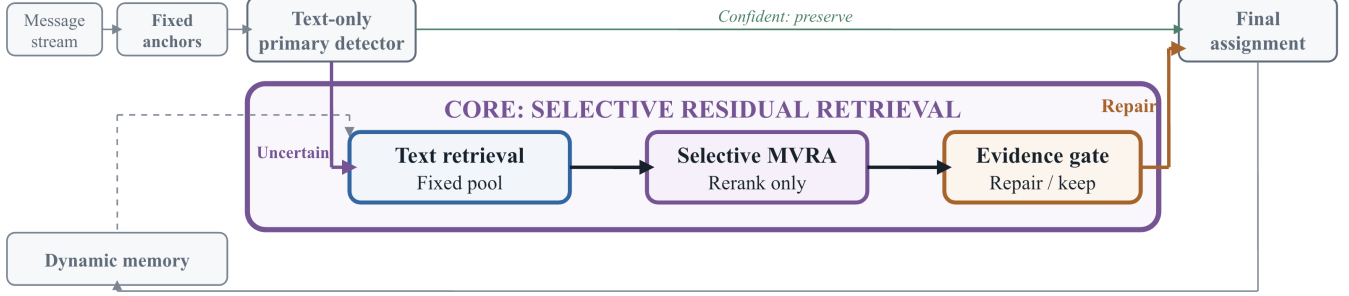


Fig. 1. Focused overview of MvSED. Uncertain primary decisions follow the emphasized residual path: text retrieval fixes the candidate pool, selective MVRA reranks it, and the evidence gate repairs or keeps the assignment. Confident decisions bypass the core.

ing [12, 5]. Query rewriting can improve recall but may drift from the original intent [13]. Multi-view learning likewise requires explicit control of noisy and redundant views [14, 15]. MvSED combines these observations by separating candidate generation, auxiliary reranking, and repair acceptance.

3. METHOD

Let $\{m_j\}_{j=1}^N$ denote the messages in block t and $\mathcal{M}_{t-1}$ the event memory accumulated from previous blocks. The task assigns an open-set event identifier to every message. For policy π , the deployment objective is

$$\min_{\pi} T_{e2e}(\pi) \quad \text{s.t.} \quad Q(\pi) \geq Q_{\text{base}} - \epsilon, \quad (1)$$

where T_{e2e} is the time until a block is committed to memory and Q is clustering quality. We report token volume separately rather than treating it as a monotonic latency surrogate. Figure 1 emphasizes the sole residual path: an uncertain primary decision triggers text retrieval, fixed-pool selective MVRA, and evidence-gated repair; a confident decision bypasses this core. Inference never accesses task labels, although the pretrained LLM contributes learned prior knowledge.

3.1. Protocol-aligned anchors and memory

We adopt RagSEDE’s single-stage anchor protocol to keep the inference units identical across methods. Sentence-BERT embeddings [16] are grouped using a fixed cosine threshold τ_a . We denote the anchor capacity by B_a , the representative-message budget by R , and the representativeness–redundancy trade-off by ρ . These quantities are selected during development and then frozen. No second-stage anchor reclustering is applied. Figure 2 illustrates the chronological construction and representative selection.

For each event, $\mathcal{M}_{t-1}$ stores its name, keywords, linked anchors, representative messages, and time observations. A

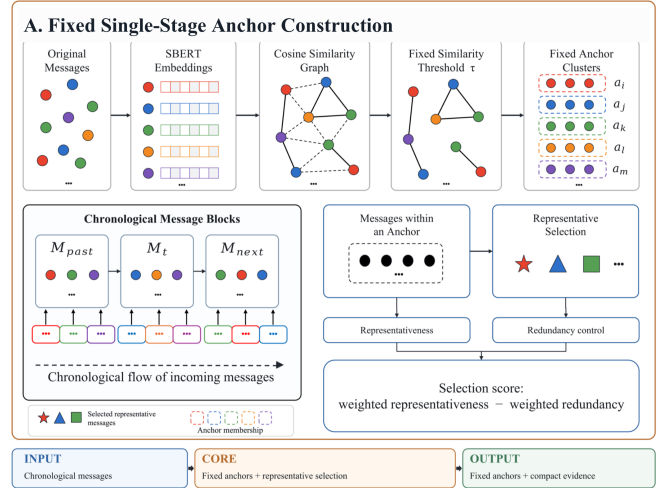


Fig. 2. Shared fixed-anchor protocol used by both methods. Chronological messages are embedded and grouped once with a fixed threshold; representative messages retain coverage while controlling redundancy.

text-only primary detector maps anchor a_i to an existing event $e \in \mathcal{M}_{t-1}$ or to OTHERS. The final decision is used to update memory.

3.2. Stage-aware selective multi-view retrieval

The selected representatives of a_i are concatenated and normalized to form the text query $q_i^{(t)}$. Figure 3 shows how this query retrieves the only admissible candidate set

$$\mathcal{C}_i = \text{TopK}_{K_c} \left(r_t(q_i^{(t)}, e), \mathcal{M}_{t-1} \right). \quad (2)$$

where r_t is raw SBERT cosine similarity. Auxiliary views $v \in \{\text{entity, relation, time}\}$ are generated from the same anchor and cannot introduce events outside $\mathcal{C}_i$. The fixed-pool constraint limits query drift to a reranking operation.

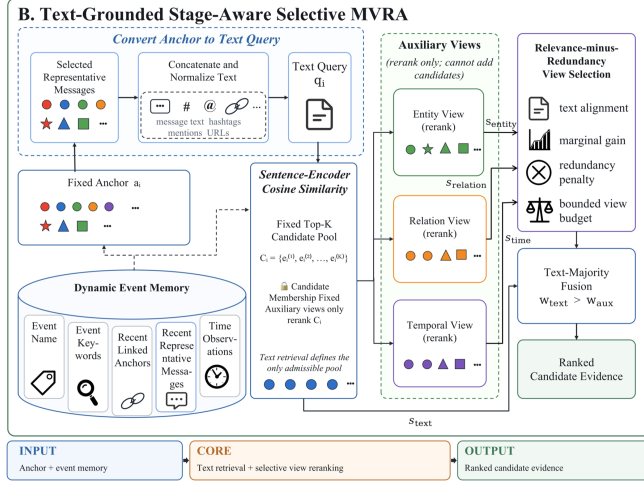


Fig. 3. Text-grounded selective MVRA. Text retrieval fixes candidate membership; entity, relation, and temporal views may only rerank that pool before text-majority fusion.

Let $\tilde{s}_v$ be view v 's normalized score vector over $\mathcal{C}_i$. Text is always retained, and at most $L - 1$ auxiliary views are selected. For auxiliary view v , define its centered ranking-profile alignment A_v to text, redundancy R_v , score dispersion σ_v , and top-candidate consistency c_v by

$$A_v = \text{corr}(\tilde{s}_v, \tilde{s}_t), \quad R_v = \max_{u \in S_a} |\text{corr}(\tilde{s}_v, \tilde{s}_u)|, \quad (3)$$

$$g(v) = \sigma_v [c_v \max(0, A_v) - \lambda R_v].$$

A view is retained only if $A_v \geq \theta_{\text{align}}$ and $g(v) \geq \theta_{\text{gain}}$. Entity and relation views provide structural support. Time may resolve a ranking tie but cannot independently justify a repair. Candidate scores are fused with a text majority,

$$s_f(e) = \omega_t \tilde{s}_t(e) + (1 - \omega_t) \sum_{v \in S_a} \alpha_v \tilde{s}_v(e), \quad (4)$$

where $\alpha_v = g(v) / \sum_{u \in S_a} g(u)$; if S_a is empty, $s_f = \tilde{s}_t$. The asymmetric weight $\omega_t > 1 - \omega_t$ preserves text as primary evidence.

3.3. Uncertainty-gated residual repair

The multi-view path is invoked only when the primary detector returns *OTHERS* or *Confusion*. Figure 4 summarizes this residual path. Let e^* and e° denote the highest-scoring candidate and its runner-up, respectively. The gate accepts a reassignment to e^* if

$$G_i = [n_{\text{view}} \geq \eta_{\text{view}}] \wedge [n_{\text{struct}} \geq \eta_{\text{struct}}] \wedge [r_t(q_i^{(t)}, e^*) \geq \theta_{\text{text}}] \wedge [s_f(e^*) \geq \theta_{\text{fuse}}] \wedge [s_f(e^*) - s_f(e^\circ) \geq \theta_{\text{margin}}]. \quad (5)$$

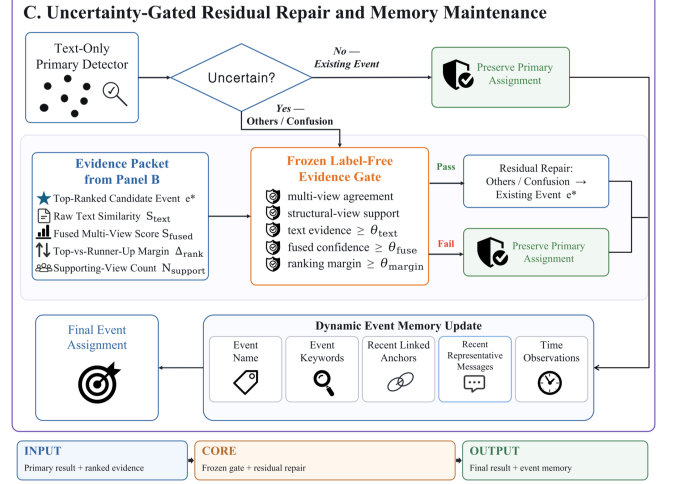


Fig. 4. Uncertainty-gated residual repair. Confident primary decisions bypass repair; uncertain decisions are changed only when the frozen evidence gate passes, after which event memory is updated.

View v supports e when e lies in its top- K_s ranking over $\mathcal{C}_i$. Here, n_{view} counts gate-eligible text, entity, and relation support, whereas n_{struct} counts entity and relation support only; time is excluded from both counts. A failed gate restores the primary output, whereas a passed gate deterministically assigns e^* ; the reported method issues no second decision-stage LLM call. Gate parameters are selected under the development protocol and frozen before testing. Held-out labels never enter inference and are consulted only post hoc for evaluation.

4. EXPERIMENTS

4.1. Experimental setup

Data and split. Event2012 contains 68,841 English tweets from 503 events collected over four weeks [17]; Event2018 contains 64,516 French tweets from 257 events over 23 days [18]. Following the incremental protocol, we evaluate 21 Event2012 blocks and 16 Event2018 blocks. Event2012 M1–M6 are used for method development. View-selection and gate parameters are then frozen for Event2012 M7–M21 and used unchanged on Event2018; only the pre-specified dataset encoder and fixed anchor threshold differ.

Models and controls. Event2012/Event2018 use all-MiniLM-L6-v2/multilingual DistilUSE. Within each ablation panel, all rows share the dataset, blocks, encoder, DeepSeek-R1-Distill-Qwen-32B endpoint, primary prompt, decoding/retry settings, fixed run seed, and evaluator; only the named module changes. Runs use one RTX 4090 (24 GB). **Metrics and timing.** The same evaluator macro-averages NMI, AMI, and ARI [19] over identical blocks within a

Table 1. Quality comparison; HISEvent[†] is non-matched [4], with values from [6].

Method	Event2012			Event2018		
	NMI	AMI	ARI	NMI	AMI	ARI
HISEvent [†]	0.80200	0.79200	0.61133	0.75375	0.74688	0.60375
RagSEDE	0.89637	0.88610	0.78386	0.75418	0.74012	0.61094
MvSED	0.89660	0.88615	0.78455	0.76400	0.75284	0.66762

Table 2. Paired efficiency and controlled ablations (R/M: RagSEDE/MvSED; E2E h includes initialization; Proc. h excludes it; Time red.: aggregate vs. RagSEDE; bold: panel best).

Data	End-to-end latency			LLM calls		Failed attempts	
	R/M (h)	Reduction	Speed-up	R/M	Reduction	R/M	Reduction
Event2012	11.15/9.89	11.3%	1.13×	2702/2668	1.3%	128/71	44.5%
Event2018	23.22/18.04	22.3%	1.29×	4749/4468	5.9%	123/57	53.7%
Combined	34.37/27.93	18.7%	1.23×	7451/7136	4.2%	251/128	49.0%
<i>Controlled Event2012 ablation (M7–M21; equal anchor counts)</i>							
Method	NMI	AMI	ARI	Proc. h	Calls	Failures	Time red.
RagSEDE	0.89593	0.88557	0.78158	11.02	2702	128	–
MVRA	0.89470	0.88428	0.78443	10.76	2741	51	2.41%
NoTime	0.89224	0.88167	0.77369	11.65	2641	70	–5.73%
MvSED	0.89306	0.88199	0.77432	9.78	2668	71	11.26%
<i>Aligned Event2018 ablation (M1–M16)</i>							
Method	NMI	AMI	ARI	Proc. h	Calls	Failures	Time red.
RagSEDE	0.75418	0.74012	0.61094	23.12	4749	123	–
MVRA	0.77229	0.76044	0.65022	20.46	4753	80	11.49%
NoTime	0.75317	0.74126	0.63830	17.06	4467	63	26.18%
MvSED	0.76400	0.75284	0.66762	17.93	4468	57	22.45%

panel. End-to-end hours include initialization and form the primary endpoint; Proc. h excludes initialization. Calls, failures, LLM time, tokens, and peak memory are diagnostics. **Anchor verification.** Event2018 ordered-anchor hashes match. Event2012 ablation rows match blockwise anchor counts, but legacy files omit some hashes; we do not claim full hash-level verification for that panel.

Frozen implementation. Anchor thresholds are 0.40 and 0.30 for Event2012 and Event2018. Each anchor contains at most 100 messages and three representatives. We set $K_c = 8$, $L = 3$, text/auxiliary weights to 0.65/0.35, alignment/gain thresholds to 0.05/0.015, and $\lambda = 0.35$. The gate requires two views, one structural view, text/fused scores of 0.45/0.40, and a 0.03 margin. Generation permits 2048 tokens, 30 attempts, and a 15-s retry delay. All values are frozen during testing; code and aggregate results are public.

4.2. Controlled quality and efficiency

Table 1 reports the controlled RagSEDE comparison and HISEvent [4]. HISEvent values are averages from RagSEDE’s published block table [6], not a controlled rerun. On held-out Event2012, the three-seed MvSED–RagSEDE changes are +0.00023, +0.00005, and +0.00069 for NMI, AMI, and ARI. Event2018 has favorable fixed-run changes of 0.0098, 0.0127, and 0.0567, winning 10, 11, and 11 of 16 blocks. All block-bootstrap 95% intervals include zero.

The upper panel of Table 2 shows that MvSED is faster on 14/15 Event2012 and 16/16 Event2018 blocks; bootstrap latency-reduction intervals are 6.9–15.8% and 18.3–26.6%. Calls fall by 1.3% and 5.9%, and failures by 44.5% and

53.7%. On Event2018, 5.18 h less end-to-end time coincides with 5.26 h less LLM time and 66 fewer failures. Retries add provider waiting, backoff, and reissuing to the serial path; retrieval and embedding partly offset the saving. This is aggregate attribution, not block-level causality.

4.3. Controlled ablation

The lower panels of Table 2 test each stage under controlled anchor counts. **Reranking only:** MVRA changes Event2012 metrics by at most 0.0029 and improves the three Event2018 metrics by 0.0181, 0.0203, and 0.0393. Time falls by 2.41%/11.49%. **Removing time:** NoTime is slower by 5.73% on Event2012 and faster by 26.18% on Event2018. **Full path:** time falls by 11.26%/22.45% and failures from 128/123 to 71/57. Event2012 tokens (M) are 1.789, 2.149, 2.071, and 2.077 for RagSEDE, MVRA, NoTime, and full. Event2018 has a 21.65% paired mean time reduction (CI: 17.26–25.93%; 16/16 wins; $p = .000031$). Thus we claim consistent latency reduction, not universal quality gains.

5. CONCLUSION

We presented MvSED: a text-grounded residual path with fixed candidates, selective reranking, and evidence-gated repair. Event2012 metric changes remain below 0.001; Event2018 shows favorable observed changes of 0.0098, 0.0127, and 0.0567. Runtime falls by 11.3% and 22.3% (18.7% combined, 1.23×). Event2018 uses 11.9% more tokens but fewer calls, failures, and 24.5% less LLM time; we therefore claim deployment-specific latency reduction, not lower cost or universal efficiency. Quality intervals include zero; labels serve only development and post-hoc evaluation. **AI assistance:** OpenAI Codex assisted language editing, Figs. 1–4 layout, and release packaging; authors designed the method and experiments and verified all content.

6. REFERENCES

- [1] Mahmud Hasan, Mehmet A. Orgun, and Rolf Schwitter, “Real-time event detection from the twitter data stream using the twitternews+ framework,” *Information Processing & Management*, vol. 56, no. 3, pp. 1146–1165, 2019.
- [2] Quanzhi Li, Yang Chao, Dong Li, Yao Lu, and Chi Zhang, “Event detection from social media stream: Methods, datasets and opportunities,” in *Proc. IEEE Big Data*, 2022, pp. 3509–3516.
- [3] Yuwei Cao, Hao Peng, Jia Wu, Yingdong Dou, Jianxin Li, and Philip S. Yu, “Knowledge-preserving incremental social event detection via heterogeneous GNNs,” in *Proc. The Web Conference*, 2021, pp. 3383–3395.
- [4] Yuwei Cao, Hao Peng, Zhengtao Yu, and Philip S. Yu, “Hierarchical and incremental structural entropy minimization for unsupervised social event detection,” in *Proc. AAAI*, 2024, vol. 38, pp. 8255–8264.
- [5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [6] Qitong Liu, Hao Peng, Zuchen Li, Xihang Meng, Ziyu Yang, Jiting Li, Li Sun, and Philip S. Yu, “Effective and unsupervised social event detection and evolution via RAG and structural entropy,” in *Proc. ACM Web Conference (WWW)*, 2026.
- [7] Minh-Son Dao, Anis Kasem, Hiroshi Hasegawa, Huynh Thi Thanh Binh Le, Akira Suzuki, and Thanh Thuy Nguyen, “Unsupervised social media events clustering using user-centric parallel split-n-merge algorithms,” in *Proc. IEEE ICASSP*, 2014, pp. 4798–4802.
- [8] Jianhua Yin and Jianyong Wang, “A dirichlet multinomial mixture model-based approach for short text clustering,” in *Proc. ACM SIGKDD*, 2014, pp. 233–242.
- [9] Jiaqian Ren, Lei Jiang, Hao Peng, Yuwei Cao, Jia Wu, Philip S. Yu, and Lifang He, “From known to unknown: Quality-aware self-improving graph neural network for open set social event detection,” in *Proc. ACM CIKM*, 2022, pp. 1696–1705.
- [10] Xiaoyan Yu, Jiaqian Ren, Lei Jiang, Hao Peng, Zhifeng Hao, Li Sun, Kun Peng, Liehuang Zhu, and Philip S. Yu, “PromptSED: An evolving topic-enhanced prompting framework for incremental social event detection,” *Neural Networks*, vol. 191, pp. 107772, 2025.
- [11] Zitai Qiu, Congbo Ma, Jia Wu, and Jian Yang, “Text is all you need: LLM-enhanced incremental social event detection,” in *Proc. ACL*, 2025, pp. 4666–4680.
- [12] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih, “Dense passage retrieval for open-domain question answering,” in *Proc. EMNLP*, 2020, pp. 6769–6781.
- [13] Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan, “Query rewriting in retrieval-augmented large language models,” in *Proc. EMNLP*, 2023, pp. 5303–5315.
- [14] Chang Xu, Dacheng Tao, and Chao Xu, “A survey on multi-view learning,” *arXiv preprint arXiv:1304.5634*, 2013.
- [15] Guowen Chao, Shiliang Sun, and Jian Bi, “A survey on multi-view clustering,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 1, pp. 567–589, 2022.
- [16] Nils Reimers and Iryna Gurevych, “Sentence-BERT: Sentence embeddings using siamese BERT-networks,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.
- [17] Andrew J. McMinn, Yashar Moshfeghi, and Joemon M. Jose, “Building a large-scale corpus for evaluating event detection on twitter,” in *Proc. ACM CIKM*, 2013, pp. 409–418.
- [18] Béatrice Mazoyer, Julia Cagé, Nicolas Hervé, and Céline Hudelot, “A french corpus for event detection on twitter,” in *Proc. LREC*, 2020, pp. 6220–6227.
- [19] Nguyen Xuan Vinh, Julien Epps, and James Bailey, “Information theoretic measures for clusterings comparison: Is a correction for chance necessary?,” in *Proc. ICML*, 2009, pp. 1073–1080.